

Frame will OR the tag-returned error code with 0xB0 before sending it back. For example, if the address offset or data length specified in the Parameter field is incorrect, or the requested read length exceeds the tag's memory area, the tag will return EPC Gen2 error code 0x03 (Memory Overrun). In this case, the Notification Frame will return error code 0xB3 along with the PC+EPC of the tag being operated, as shown below:

Type: 0x01
 Command: 0xFF
 Parameter Length: 0x0010
 Parameter: 0xB3
 URL Length: 0x0E
 PC: 0x3400
 EPC: 0x30751FEB705C5904E3D50D70
 Checksum: 0x12

Header	Type	Command	PL (MSB)	PL (LSB)	Parameter	UL	PC (MSB)	PC (LSB)
BB	01	FF	00	10	B3	0E	34	00
EPC (MSB)								
30	75	1F	EB	70	5C	59	04	E3
EPC (LSB)			Checksum			End		
70			12			7E		

2.10 Lock the Lock Tag Data Storage Area

2.10.1 Command Frame

For a single tag, lock or unlock the tag's memory area. This command should be preceded by setting the Select parameter to choose the specific tag for the Lock operation. For example, to lock the Access Password, use the following command:

Type: 0x00
 Command: 0x82
 Parameter Length: 0x0007
 Access Password: 0x0000FFFF
 Lock Operation-LD: 0x020080
 Checksum: 0x09

Header	Type	Command	PL (MSB)	PL (LSB)	AP (MSB)
BB	00	82	00	07	00 00 FF
AP (LSB)		LD (MSB)	LD (LSB)	Checksum	End
FF		02	00	80	09 7E

LD's upper 4 bits are Reserved, while the remaining 20 bits constitute the Lock operation payload, which consists of Mask and Action—10 bits each, ordered from MSB to LSB. For detailed information, refer to EPC Gen2 standard v1.2.0, Section 6.3.2.11.3.5.

Mask serves as a bit-mask; only actions whose corresponding Mask bit is 1 are valid. Each memory area's Action occupies 2 bits (00 – 11), representing in sequence: Open,

Permanently Open, Lock, Permanently Lock.

For example, if the Kill Mask is 2 bits 00, the Kill Action is ignored regardless of its value. When the Kill Mask is 2 bits 10 and the Kill Action is 2 bits 10, the Kill Password is locked (non-permanent lock) and can be read or written only with a valid Access Password.

The meaning of each Mask and Action bit is shown below.



Masks and Associated Action Fields

	Kill pwd		Access pwd		EPC memory		TID memory		User memory	
	19	18	17	16	15	14	13	12	11	10
<i>Mask</i>	skip/ write	skip/ write	skip/ write	skip/ write	skip/ write	skip/ write	skip/ write	skip/ write	skip/ write	skip/ write
	9	8	7	6	5	4	3	2	1	0
<i>Action</i>	pwd read/ write	perma lock	pwd read/ write	perma lock	pwd write	perma lock	pwd write	perma lock	pwd write	perma lock

pwd-write	permalock	Description
0	0	Associated memory bank is writeable from either the open or secured states.
0	1	Associated memory bank is permanently writeable from either the open or secured states and may never be locked.
1	0	Associated memory bank is writeable from the secured state but not from the open state.
1	1	Associated memory bank is not writeable from any state.
pwd-read/write	permalock	Description
0	0	Associated password location is readable and writeable from either the open or secured states.
0	1	Associated password location is permanently readable and writeable from either the open or secured states and may never be locked.
1	0	Associated password location is readable and writeable from the secured state but not from the open state.
1	1	Associated password location is not readable or writeable from any state.

2.10.2 Notification Frame

If the Lock command is executed correctly and the tag's response is valid, the Notification Frame is as follows:

Type: 0x01
 Command: 0x82
 Parameter Length: 0x0010
 URL Length: 0x0E
 PC: 0x3400
 EPC: 0x30751FEB705C5904E3D50D70
 Data: 0x00 (Execution successful)
 Checksum: 0xE2

Header	Type	Command	PL (MSB)	PL (LSB)	UL	PC (MSB)	PC (LSB)
BB	01	82	00	10	0E	34	00
EPC (MSB)							
30	75	1F	EB	70	5C	59	04
						E3	D5
							0D

EPC (LSB)	Parameter	Checksum	End
70	00	E2	7E

If the tag is not within the field or the specified EPC code is incorrect, error code 0x13 will be returned as follows:

Type: 0x01
Command: 0xFF
Parameter Length: 0x0001
Parameter: 0x13
Checksum: 0x14

Header	Type	Command	PL (MSB)	PL (LSB)	Parameter	Checksum	End
BB	01	FF	00	01	13	14	7E

If the Access Password is incorrect, error code 0x16 will be returned, along with the PC+EPC of the tag being operated, as follows:

Type: 0x01
Command: 0xFF
Parameter Length: 0x0010
Parameter: 0x16
URL Length: 0x0E
PC: 0x3400
EPC: 0x30751FEB705C5904E3D50D70
Checksum: 0x75

Header	Type	Command	PL (MSB)	PL (LSB)	Parameter	UL	PC (MSB)	PC (LSB)
BB	01	FF	00	10	16	0E	34	00
EPC (MSB)								
30	75	1F	EB	70	5C	59	04	E3
EPC (LSB)			Checksum			End		
70			75			7E		

If the operated tag returns an error code defined by the EPC Gen2 protocol, note that only the lower 4 bits of EPC Gen2 error codes are valid. Consequently, the Notification Frame will OR the tag-returned error code with 0xC0 before sending it back. For example, if the address offset or data length specified in the Parameter field is incorrect, or the requested read length exceeds the tag's memory area, the tag will return EPC Gen2 error code 0x04 (Memory Overrun). In this case, the Notification Frame will return error code 0xC4 along with the PC+EPC of the tag being operated, as shown below:

Type: 0x01
Command: 0xFF
Parameter Length: 0x0010
Parameter: 0xC4
URL Length: 0x0E
PC: 0x3400
EPC: 0x30751FEB705C5904E3D50D70
Checksum: 0x23